

Bank Marketing Project

Group 9

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Problem Statement

Data Collection

The dataset used in this analysis comes from the Bank Marketing dataset hosted on the UCI Machine Learning Repository. It contains information on customer demographic attributes, their financial and social characteristics, details of previous marketing contacts, and the outcome of a direct marketing campaign.

Before specifying and fitting any statistical model, it is essential to understand the structure, quality, and behavior of the data. To this end, we conduct an exploratory data analysis (EDA) to examine variable types, detect missing or unknown categories, assess the distribution of numerical predictors, and inspect potential relationships between predictors and the response variable.

This initial exploration guides the cleaning and preprocessing decisions required for logistic regression, ensures that modeling assumptions are not violated early on, and helps identify which variables may contribute meaningfully to predicting the subscription outcome. Below, we provide some exploratory data analysis and a first look at the structure of the dataset. This includes examining the variables, their distributions, and any initial patterns that may inform subsequent modeling decisions.

The dataset comprises observations and variables.

Initial Data Inspection

Before performing any detailed analysis, it is important to gain a preliminary understanding of the dataset's structure. In this section, we inspect the column names and their corresponding data types to identify which variables are numeric, categorical, or character. This initial inspection provides a foundation for subsequent exploratory data analysis and helps guide decisions on preprocessing, encoding, and handling special or unknown values. The description for each column, its data type, number of unique values and number of missing values are shown below:

Table 1: Column Description

Col_name	Type	Unique_Class	Num_missing
age	integer	67	0
job	character	12	0
marital	character	4	0
education	character	8	0
default	character	3	0
housing	character	3	0
loan	character	3	0
contact	character	2	0
month	character	10	0

Col_name	Type	Unique_Class	Num_missing
day_of_week	character	5	0
duration	integer	828	0
campaign	integer	25	0
pdays	integer	21	0
previous	integer	7	0
poutcome	character	3	0
emp.var.rate	numeric	10	0
cons.price.idx	numeric	26	0
cons.conf.idx	numeric	26	0
euribor3m	numeric	234	0
nr.employed	numeric	11	0
y	character	2	0

Exploratory Data Analysis

Univariate EDA

Bivariate EDA

Multivariate EDA

Data Cleaning and Feature Preparation

Handling “Unknown” Values

Handling Special Numeric Values

Outlier Treatment

Encoding Categorical Variables

Feature Scaling

Model Specification

Model Fitting

Model Validation and Criticism

Interpretation of Coefficients

Prediction